IndieSemiC ISC-SX1262-B LoRa Module Pinout

Complete SMD-16 Pinout & Connection Guide for Seeed Studio XIAO ESP32C6

Semtech SX1262 Core | 868 MHz Sub-GHz | +22 dBm High Power

IndieSemiC ISC-SX1262-B (16-Pin Stamp-Hole Package)

LEFT SIDE (PINS 1 → 8)

Pin 1	ANT	868MHz Antenna / IPEX	RF
Pin 2	GND	Ground Rail	GND
Pin 3	3.3V	+3.3V Power Rail	3V3
Pin 4	RESET	XIAO RST Pad (Sync)	RESET
Pin 5	TXEN	Not Connected (Floating)	NC
Pin 6	DIO1	XIAO D2 (GPIO 2)	IRQ
Pin 7	DIO2	Not Connected (Internal RF SW)	NC
Pin 8	DIO3	Not Connected (Internal TCXO)	NC

RIGHT SIDE (PINS 16 → 9)

Pin 16	GND	Ground Rail	GND
Pin 15	NSS	XIAO D3 (GPIO 21)	SPI CS
Pin 14	MOSI	XIAO D10 (GPIO 18)	SPI MOSI
Pin 13	MISO	XIAO D9 (GPIO 20)	SPI MISO
Pin 12	SCK	XIAO D8 (GPIO 19)	SPI SCK
Pin 11	RXEN	Not Connected (Floating)	NC
Pin 10	BUSY	XIAO D1 (GPIO 1)	BUSY
Pin 9	GND	Ground Rail	GND

Official ISC-SX1262-B Pinout & Wiring Table

Pin #	Pin Name	Datasheet Function	Connect to XIAO ESP32C6	Status
1	ANT	RF Antenna Output (50Ω)	Attach 865-868MHz Antenna via IPEX/SMA Pigtail	**Required (RF)**
2	GND	Module Ground Reference	Common GND Bus	**Required (Power)**
3	3.3V	DC Power Input (2.7V - 3.6V)	+3.3V Rail (from AP2112K LDO)	**Required (Power)**
4	RESET	Transceiver Hardware Reset (Active LOW)	Connect to XIAO **RST Pad** (Hardware Sync)	**Required (Bypass)**
5	TXEN	RF Switch Transmit Control	Leave Disconnected (Floating)	*Not Connected*
6	DIO1	Digital Interrupt (Packet Rx/Tx Done)	XIAO **D2** (GPIO 2)	**Required (Signal)**
7	DIO2	Digital IO 2 / RF Switch Driver	Leave Disconnected (Controlled Internally)	*Not Connected*
8	DIO3	Digital IO 3 / TCXO Voltage Feed	Leave Disconnected (Controlled Internally)	*Not Connected*
9	GND	Module Ground Reference	Common GND Bus	**Required (Power)**
10	BUSY	Transceiver Busy Status Line	XIAO **D1** (GPIO 1)	**Required (Signal)**
11	RXEN	RF Switch Receive Control	Leave Disconnected (Floating)	*Not Connected*
12	SCK	SPI Clock Input	XIAO **D8** (GPIO 19)	**Required (SPI)**
13	MISO	SPI Data Output (Radio → MCU)	XIAO **D9** (GPIO 20)	**Required (SPI)**
14	MOSI	SPI Data Input (MCU → Radio)	XIAO **D10** (GPIO 18)	**Required (SPI)**
15	NSS	SPI Chip Select (Active LOW)	XIAO **D3** (GPIO 21)	**Required (SPI)**
16	GND	Module Ground Reference	Common GND Bus	**Required (Power)**

Soldering & Hookup Summary:
• **Pins Connected to MCU (6 wires):** Pin 6 (DIO1) → D2, Pin 10 (BUSY) → D1, Pin 12 (SCK) → D8, Pin 13 (MISO) → D9, Pin 14 (MOSI) → D10, Pin 15 (NSS) → D3.
• **Hardware Reset Sync (1 wire):** Pin 4 (RESET) → XIAO underside RST solder pad.
• **Power Rails:** Pin 3 (3.3V) → 3.3V Rail; Pins 2, 9, 16 → Common GND plane.
• **Unused Pins (Floating):** Pins 5 (TXEN), 7 (DIO2), 8 (DIO3), and 11 (RXEN) do NOT connect to anything.
